Adeola Adegbemijo

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Applied Data Scientist

Research Data Scientist with PhD-level training and industry experience at top-scale platforms (AWS, Meta), specializing in experimentation, marketing and ads metrics, user behavior analysis, and causal inference. Proven track record of designing scalable data science solutions that measure product quality, quantify incremental lift, and influence long-term product strategy for advertising and platform products.

RESEARCH MOTIVATION

I am interested in how intelligent systems extract structure from noisy, high-dimensional data, and how experimentation, small language models, and evaluation frameworks can be used to study, improve, and align intelligent behavior in real-world settings. My work sits at the intersection of AI/ML, experimental design, and systems thinking, with an emphasis on building and empirically validating ideas.

RESEARCH & OPEN-SOURCE PROJECTS

AI-Assisted Rapid Qualitative Analysis (AI-RQA)

- Formulated the original hypothesis that small language models could approximate expert qualitative reasoning under constrained settings
- Built and deployed transformer-based NLP systems (DistilBERT, ELECTRA) to study how small language models extract structured signals from unstructured text.
- Designed evaluation frameworks comparing model outputs against expert human annotations, achieving near-human agreement.
- Investigated failure modes, bias, and privacy constraints in ML systems applied to sensitive real-world domains.
- Operationalized the system into reproducible pipelines used by non-technical researchers.

pytranscripts (Open-source Python Library)

- Co-developed an open-source library enabling local, privacy-preserving qualitative analysis using machine learning models.
- Focused on usability, interpretability, and reproducibility of ML-assisted analysis workflows.
- Designed the library to support rapid iteration and experimentation by independent researchers.

TECHNICAL PROFICIENCIES

Programming Language/Data Visualization: Python, SQL, Sagemaker, S3, Mathematica, SAS, QuickSight
Statistics & Modeling: Regression, Classification, Bayesian Methods, Forecasting, Optimization, Hierarchical reconciliation
Research, Data Science & Modeling: Machine Learning & Heuristic Algorithms, ML Engineering, University Teaching, Experimentation Research, A/B Testing, Synthetic Difference-in-Differences, Randomization, Placebo Tests
Optimization & OR: Gurobi, CPLEX
AI/Machine Learning: LLMs, Agentic AI, K-NN, Neural Networks, SVM, Decision Trees, Regression, Naïve Bayes, Clustering, NLP

SKILLS

Coursework: Operations Research, Systems Science, Complex Design Systems, Advanced Applied Econometrics, Computational Tools, Probability & Statistics, Applied Multivariate Data Analysis, Advanced Network Science, Bio-Inspired Computing, Machine Learning
Certification: Amazon MLU Agentic AI application: From Concept to Code

RESEARCH INTERESTS

Computational Social Science, Machine Learning & Artificial Intelligence, Industry 4.0, Complex Design Systems, Network Science, Data Science, Natural Language Processing (NLP), Small Language Modelling (SLM)

PROFESSIONAL EXPERIENCE

AWS Marketing Data Scientist – Amazon Web Services
Experimentation Engineering **September 2024 – Present**

- Design and execute multi-region, multi-cell incrementality experiments across paid media channels (content syndication, programmatic display) spanning EMEA, NAMER, and Partner segments, using randomized geo-level and account-level test designs to estimate causal marketing impact on full-funnel outcomes (MALs, MQLs, signups, FTSAs, pipeline, and revenue)
- Develop and maintain reusable causal inference frameworks for synthetic difference-indifference, panel OLS estimation with two-way fixed effects (entity + time), clustered standard errors, and pairwise treatment-group comparisons — enabling scalable, reproducible incrementality measurement across experiment rounds
- Lead post-experiment analysis including lead-level conversion modeling with longitudinal follow-up windows (4–26 weeks), heterogeneous treatment effect estimation across TALs and segments, and statistical testing for differential platform effectiveness
- Own the end-to-end experiment lifecycle: randomization design, population equivalence validation, treatment indicator construction, descriptive diagnostics, regression analysis, and results interpretation for leadership decision-making on multi-million-dollar marketing investments
- Architecting an AI-powered experimentation agent to automate and improve the always-on global incrementality measurement pipeline — orchestrating data validation, causal estimation, result interpretation, and reporting across regions and channels with minimal manual intervention
- Built an ensemble forecasting model integrating ML-based learners that reduced Monthly Business Review and Annual goal-setting MAPE by 15% relative to existing baseline, with hierarchical reconciliation across heterogeneity splits for consistent cross-functional reporting
- Designed an unsupervised ML-powered profiling intelligence system using clustering and embedding-based similarity to optimize strategist-to-opportunity mapping for deals ≥$1M in potential revenue, accelerating conversion cycles
- Built a supervised ML prediction model layered on revenue causal estimates to determine client engagement outcomes for Enterprise strategists, achieving 71% F1-score
- Drove the development and deployment of deep learning models that enhance predictive accuracy through advanced pattern recognition and semantic understanding, directly improving forecasting and decision intelligence
- Leverage AI-assisted tooling and LLM-based workflows to accelerate experiment analysis pipelines, automate reporting, and scale measurement infrastructure across teams
- Partner with Product, Design, and Engineering to integrate incrementality models, AI-driven insights, and measurement frameworks into reporting workflows and investment decision-making systems

Graduate Researcher – State University of New York at Binghamton
AI/ML Application & Implementation to Process Improvement **August 2023 – September 2025**

- Built and evaluated NLP systems for rapid qualitative and quantitative analysis of real-world reports related to medical processes interventions through implementation science.
- Built and deployed AI-RQA (AI-Assisted Rapid Qualitative Analysis), leveraging Small Language Models (DistilBERT, ELECTRA) to automate qualitative healthcare coding, achieving near-human agreement with expert annotators
- Influenced methodological design by partnering with implementation scientists to align model outputs with established qualitative frameworks.
- Operationalized ethical, social, and data privacy implications of AI-assisted qualitative research, producing guidelines for responsible deployment of SLMs in healthcare contexts
- Co-created open-source tools (pytranscripts Python library and a Streamlit interface) to operationalize AI-RQA, enabling non-technical researchers to analyze sensitive data locally while safeguarding privacy

AWS Data Scientist Internship – Amazon Inc.

Experimentation and Bayesian testing **June 2023 – September 2023**

- Built statistical metrics and standardizing production and reporting across the six teams to improve testing program
- Drove a proof of concept that scaled experimentation velocity and improved test execution.
- Led the development of Bayesian and Sequential Probability Ratio test models to optimize test duration and increase complexity of analysis to the forefront of academic and industry standards. This improvement directly led to a 20% increase in the number of experiments carried out by the program yearly
- Recommended and ran appropriate A/B tests (or other) to measure the impact of site changes
- Worked cross-functionally with Design, Product Management, and Engineering to incorporate user-centered research at all stages of the product cycle.

Graduate Researcher – State University of New York at Binghamton

Complex Systems Modeling **August 2022 – May 2023**

- Investigated SEIR (epidemic) Network models and predictive analysis.
- Conducted research to develop improved iteration of network-based compartmental modeling using evolutionary game theory, replicator dynamics, machine learning algorithm, and mean-field approximation.
- Authored research paper titled 'Epidemic Dynamics of Network in Determining Optimal Human Mobility.'

Industry 4.0 Systems Performance & Maturity benchmarking **August 2021 – May 2022**

- Designed a system architecture for the information management system supporting the assessment in a cloud-based platform
- Built a prototype system for beta testing the industry 4.0 maturity assessment recommendation
- Applied qualitative and quantitative research questions into a survey plan to maximize insights
- Implemented an approach to statistically measure and perform computational analysis for manufacturing companies based on Industry 4.0 benchmarking score

Teaching Assistant (Probabilistic Systems) – State University of New York at Binghamton

Applied statistics, experimentation, and hypothesis testing for engineers **January 2023 – May 2023**

- Led instruction and evaluation for applied statistics and experimentation courses towards the following learning outcomes;
  - Performing descriptive statistics,
  - Understanding probability theory and compare common types of continuous and discrete distributions,
  - Conductingparament estimation (point estimation, MLE, confidence intervals),
  - Performing hypothesis tests for single and two samples.

Quantitative UX Research Intern – Facebook/Meta Inc Platforms

Content ranking and enforcement metrics improvement **May 2022 – August 2022**

- Influenced product decisions by developing recommendations using ranking analysis and sigmoid function in Python to recalibrate the current accuracy metrics to reduce false positive and increase true positive metrics for harmful contents
- Developed and ratified a golden set model for evaluating metrics used to improve the efficiency of human moderators for enforcement of violating content, on the community of platforms with senior stakeholders
- Proactively designed research to understand metric performance versus value representation
- Collected and analyzed value representation data through surveys and log analysis using Qualtrics and Diaquery/SQL

EDUCATION

State University of New York at Binghamton — Expected September 2026
Ph.D. Systems Science & Industrial Engineering — GPA – 3.92

Claremont Graduate University, CA — December 2020
M.A International Studies/Computational Analytics — GPA – 3.85

Obafemi Awolowo University, Nigeria — October 2016
B.A. English Language

PUBLICATIONS

- **Adegbemijo, A**. et al. 2025. AI-assisted Rapid Qualitative Analysis for Implementation Science: Shifting from IF to HOW Enhancing the speed of qualitative analysis for implementation scientists using a small language model (Preprint). JMIR. (Jul. 2025). DOI:https://doi.org/10.2196/preprints.81149
- **Adegbemijo, A.,** Jesso, S. T., & Maw, A. (2024, December 9). Developing and validating an AI-assisted approach to rapid qualitative analysis for Implementation Science Practitioners. AcademyHealth - 17Th Annual Conference on the Science of Dissemination and Implementation. https://academyhealth.confex.com/academyhealth/2024di/meetingapp.cgi/Paper/69315
- Tulk Jesso, S., Greene, C., Zhang, S., Booth, A., DiFabio, M., Babalola, G., **Adegbemijo, A**., & Sarkar, S. (2024). On the potential for human-centered, cognitively inspired AI to bridge the gap between optimism and reality for autonomous robotics in healthcare: a respectful critique. Proceedings of the International Symposium on Human Factors and Ergonomics in Health Care, 13(1), 106-112. https://doi.org/10.1177/2327857924131080
- **A. Adegbemijo,** O. Olatunde**,** C. Cheng, A. Ademola **'**Epidemic Dynamics of Network in Determining Optimal Human Mobility' (2023). Accepted at the INFORMS 2023 Conference
- **A. Adegbemijo**, S.W. Yoon, Y. Wang, "Industry 4.0 Maturity Level Assessment for Small and Medium-sized Manufacturing Enterprises", Proceedings of the IISE Annual Conference & Expo (2022). [Link]
- Babalola G, Adedoyin A, **Adegbemijo A,** and Akintunde A, "Web-Based Forecasting and Interactive Visualization of Cardiovascular Diseases using Shiny App and Machine Learning Techniques", Proceedings of the IISE Annual Conference & Expo (2022). [Link]

LEADERSHIP EXPERIENCE

- Senator – Graduate African Student Association, SUNY Binghamton — August 2020 – Present
- Youth Volunteer – Excel Community Classes, Nigeria — September 2019 – Present

AFFILIATIONS & MEMBERSHIP

National Society of Black Engineers (NSBE) – Collegiate Member Institute of Industrial and Systems Engineers (IISE) – Collegiate Member Society of Women Engineers (SWE) – Collegiate Member

Institute for Operations Research and the Management Sciences (INFORMS) The Smart Energy Operations Research Laboratory (SEORL) Watson Institute for Systems Excellence (WISE)